

Solution Requirements

Date	1 July 2026
Team ID	
Project Name	Voice based concept understanding analyser
Maximum Marks	4 Marks

Solution Requirements Overview

Define Functional and Non-Functional Requirements to map exactly what your solution will do and how well it will perform. Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do), while Non-Functional Requirements (NFR) specify operational constraints, properties, and standards judging the overall execution behavior (How the system should be).

Step 1: Functional Requirements (FR)

S.NO	REQUIREMENT CATEGORY	REQUIREMENT DESCRIPTION	PRIORITY
1	Authentication	Users must log in via OAuth 2.0 institutional email single sign-on (SSO) or a secure email/password credential system protected by JWT tokens.	High
2	Authorization levels	The system enforces Role-Based Access Control (RBAC): Student role grants microphone recording and individual report access. Instructor role permits syllabus configuration and batch analytics viewing.	High
3	External interfaces	Integrates securely with the external OpenAI Whisper API for voice-to-text audio stream decoding, and syncs via API gateways with external Learning Management Systems (LMS) like Moodle.	High
4	Transactions processing	Handles transactional updates by streaming concurrent audio payloads to processing workers and ensuring absolute accuracy when recording final scoring matrices to transactional data repositories.	Medium
5	Reporting	Generates instantaneous web dashboard visual displays showcasing color-coded textual comparisons, vector match scores, missing knowledge node graphs, and downloadable multi-format report scorecards.	High

S.NO	REQUIREMENT CATEGORY	REQUIREMENT DESCRIPTION	PRIORITY
6	Business rules, etc.	A voice assessment must evaluate audio lengths between 15 seconds to 5 minutes, calculate similarity values using strict semantic weights, and flag answers containing less than 3 key industry terms as incomplete.	High
7	Compliance to laws	The platform ensures complete data transparency adhering strictly to local data privacy guidelines and GDPR frameworks; all recorded audio payloads are scrubbed of PII identifiers.	High
8	Other	Includes a quick background sound sensor feature during peripheral device setup to check noise levels prior to initializing automated scoring evaluations.	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR CATEGORY	REQUIREMENT DESCRIPTION	TARGET METRIC / ACCEPTANCE CRITERIA
1	Performance & Speed	System audio ingestion, text transcription processing, and similarity matrix calculations must serve feedback swiftly.	AI pipeline execution and report layout generation finishes in < 3.5 seconds from upload.
2	Scalability	The architecture uses cloud infrastructure backends capable of tracking growth during concurrent multi-user quiz intervals.	Horizontal cluster scaling reliably supports up to 5,000 concurrent student micro streams.
3	Security & Data Privacy	All stored profiles, user tokens, and evaluation text databases must protect confidentiality.	End-to-end data transit encrypted via TLS 1.3, and at-rest databases protected by AES-256.
4	Reliability & Availability	The assessment microservices must maintain continuous active uptime during operational testing schedules.	Guarantees a 99.9% uptime metric per month with failover infrastructure clustering.
5	Usability & Accessibility	Interfaces should adapt seamlessly across varying layouts, displaying high visual clarity for text and graph assets.	Fully complies with WCAG 2.1 AA accessibility standards and works responsively.
6	Other	The code baseline must use modular architectural separations to guarantee effortless future maintainability.	Documentation score meets > 90% codebase coverage via automated linting frameworks.